CLASS 6 — SCIENCE · HARD LEVEL CHAPTER TEST SET 2

Chapter 6: **Materials Around Us** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. After removing iron filings, adding water to the mixture and filtering separates: 1

(a) sand on the filter paper **(b)** salt on the filter paper
(c) both on the filter **(d)** nothing

2. Which of these should not be done with glass items? 1

(a) drying **(b)** washing with water
(c) heating suddenly **(d)** keeping on a shelf

3. Which statement is correct? 1

(a) all opaque materials are dull **(b)** all shiny materials are transparent
(c) a material that is opaque may be shiny **(d)** all transparent materials are soluble

4. To recover salt from the filtrate we should: 1

(a) add iron **(b)** evaporate the water
(c) add sand **(d)** use a magnet

5. Why is cloth used for making sails? 1

(a) it is heavy **(b)** it is light and catches wind
(c) it is soluble **(d)** it is transparent

6. Salt dissolves faster in hot water than in cold water. This suggests that: 1

(a) salt melts **(b)** water is a metal
(c) temperature affects dissolving **(d)** cold water is heavy

7. A designer chooses rubber for a tyre and glass for a window. These choices are based on: 1

(a) the names of the materials **(b)** the colours of the materials
(c) the smell of the materials **(d)** the properties of the materials

8. Why are bridges built using steel and concrete rather than wood only? 1

(a) colour **(b)** smell
(c) strength and durability **(d)** transparency

9. **Assertion (A):** Sand dissolves in water. 1
Reason (R): Salt dissolves in water.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** Metals generally show lustre (shine). 1
Reason (R): All metals are very soft and can be cut with a knife.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Why is it risky to use a thick glass tumbler for very hot liquid? 2

12. Why is wool used for sweaters but not for raincoats? 2

13. How would you test which of two white powders is sugar and which is chalk, without tasting? 2

14. Which property of metals allows them to be made into utensils of different shapes? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A mixture contains sand, salt, iron filings and sawdust. Outline steps to separate all four. 3

16. Two beakers contain equal volumes of water. A contains sugar and B salt (same amount). Which dissolves faster? Design a fair test. 3

17. How would you separate wax from a mixture of wax and sand? 3

OR

Why does a wooden spoon not get hot in boiling soup but a steel spoon does? 3

18. A packet is labelled 'translucent'. What does this mean and give an example. 3

OR

Explain how the properties of materials guided the choice of a pressure cooker. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** Arjun stirred a spoonful of each of the following in separate glasses of water: salt, sugar, sand, chalk powder and a few drops of oil. Salt and sugar vanished. Sand and chalk settled at the bottom. The oil floated on top. 5
(i) Which two substances are soluble in water?
(ii) Name one insoluble substance that settled.
(iii) Which substance floated on the water?
(iv) Why does oil float on water? Name one material that dissolves in water but is not seen again.

20. Explain the difference between 'object' and 'material' with five pairs, and show how one material can make many objects. 5

OR

Write a report on testing five materials for four properties and give your table of results and conclusion. 5

END OF QUESTION PAPER — Total: 40 Marks